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Induction Proofs

Video: Proof by Induction
7 min

Quiz: Proof by Induction Quiz
Started

Limits and Continuity

Proof by Induction Quiz

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Due Feb 25, 11:59 PM IST

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1.

To solidify how a proof by induction works, lets quickly go through a very simple example together.

1 / 1 point

Prove that $n < n + 1$. (For now just focus on $n \geq 0$)

Which of the following could be appropriate base cases to use:

☒ Let $n = 0$. Then it follows that $0 < 1$ for $n < n + 1$

☐ Let $n = x$, where x is some positive integer. Then $x < x + 1$.

☒ Correct

Try again
2.

Now that we have a base case, which of the following assumptions should we make?

1 / 1 point

☐ Assume that $n < n + 1$

☒ Assume that for all values up to $n = k$, that $n < n + 1$.

☒ Correct

Great! Now we must then show the inequality is true for $n = k + 1$
3.

Which of the following inductive steps finalizes our proof:

1 / 1 point

☒ We can show that $n = k + 1$ allows for $n < n + 1$ to remain true for any $k \geq 0$.
 $(k + 1) < (k + 1) + 1 \rightarrow k + 1 < k + 2 \rightarrow 1 < 2$
Thus by induction we see our statement is true for any $k \geq 0$

☐ We can show that $n = k + 1$ allows for $n < n + 1$ to remain true for any $k \geq 1$.
 $(k + 1) < (k + 1) + 1 \rightarrow k + 1 < k + 2 \rightarrow 1 < 2$

☒ Correct

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